

History of Computing

The computer as we know it today had its beginning with a 19th century English mathematics professor name Charles Babbage.

He designed the Analytical Engine and it was this design that the basic framework of the computers of today are based on. The Analytical Engine was a proposed mechanical general-purpose computer. It was first described in 1837 as the successor to Babbage's difference engine

Eras of Computers

1. Pre-Mechanical Era – Before 1450

This era saw the development of various tools like the abacus s the first calculating device in the world used by Chinese, the Antikythera mechanism in ancient Greece, and the astrolabe in the Islamic world. These devices were used for calculation, navigation, and astronomical purposes.



Antikythera



Astrolabe

2. Mechanical Era – 1450-1840

The Pascaline calculator (1642)

The Pascaline calculator, also known as the arithmetic machine or Pascaline, was an early mechanical calculator invented by the French mathematician, physicist, and philosopher Blaise Pascal in 1642.

It was one of the earliest devices designed to perform arithmetic calculations automatically. It was designed to work with the decimal system, allowing users to perform addition and subtraction with numbers ranging from 0 to 9.

The Pascaline consisted of a series of rotating wheels, each representing a digit in the decimal system. Users input numbers by turning the wheels to the desired digits. One of the notable features of the Pascaline was its automatic carrying mechanism, which allowed it to handle multi-digit numbers and perform carry-over operations during addition.

While revolutionary for its time, the Pascaline had limitations. It could only directly perform addition and subtraction only. Multiplication and division had to be performed through repeated addition or subtraction.



Pascaline Calculator

The Stepped Reckoner (1694)

The Stepped Reckoner, also known as the Leibniz calculator, is a mechanical calculator invented by the German mathematician and philosopher Gottfried Wilhelm Leibniz, was completed in 1694. It is notable for its ability to perform various arithmetic operations, including addition, subtraction, multiplication, and division.



Stepped Reckoner

Jacquard loom Machine (1804)

Before the 1800s, complex woven designs were created by hand. This was labor-intensive. As a result, patterned fabric was extremely expensive.

Joseph Marie Jacquard, a French weaver and inventor, developed the Jacquard loom in 1804-1805. This loom used a series of punched cards to control the pattern being woven in textiles. Each card corresponded to one row of the design, and the presence or absence of holes in the card controlled which threads were raised or lowered during weaving.

The Jacquard loom was revolutionary because it allowed for the automation of complex patterns in fabric weaving, greatly increasing efficiency and consistency in textile production. It is considered one of the earliest examples of programmable machines.



The Difference Engine (1822)

The Difference Engine, conceived by English mathematician and inventor Charles Babbage in the early 19th century with plans dating back to 1822, is considered one of the earliest designs for a mechanical computer.

Although Babbage never completed the construction of the Difference Engine during his lifetime, his work laid the foundation for modern computing concepts. He presented the concept of Input, Process, and Output, which is fundamental to modern computers.

Therefore, he is considered as the “**Father of the computer**”.



Charles Babbage



Difference engine

The Analytical Engine (1837 - 1871)

Charles Babbage designed and partly built the Analytical Engine in the 19th century. (He worked on it until his death in 1871). Although never completed in his time, it is generally considered to be the first computer.

Babbage first began working on automated digital devices in the 1820s. His initial project was the Difference Engine, designed to calculate mathematical tables. While working on the device, he began to imagine ways to improve it. He mainly considered generalizing its operation so that it could perform other kinds of calculations.

Funding ran out for his Difference Engine in 1833. By that time, however, Babbage had conceived of the Analytical Engine, a general-purpose computing machine.

The Analytical Engine was designed to be an automatic digital mechanical general-purpose computer controlled by programs. It would have been capable of performing any calculation set before it.

The machine consisted of four components:

1. The mill
2. The store
3. The reader
4. The printer

These components are similar to those found in modern computers, including The mill was the calculating unit, similar to the central processing unit (CPU) in a modern computer,

The store was where data were held prior to processing like memory and storage in today's computers, The reader and printer were the input and output devices.

The Analytical Engine was far more complex than any prior device. The store was to be large enough to hold 1,000 50-digit numbers. This was larger than the storage capacity of any computer built before 1960. The machine was to be steam-driven, with one attendant running it.

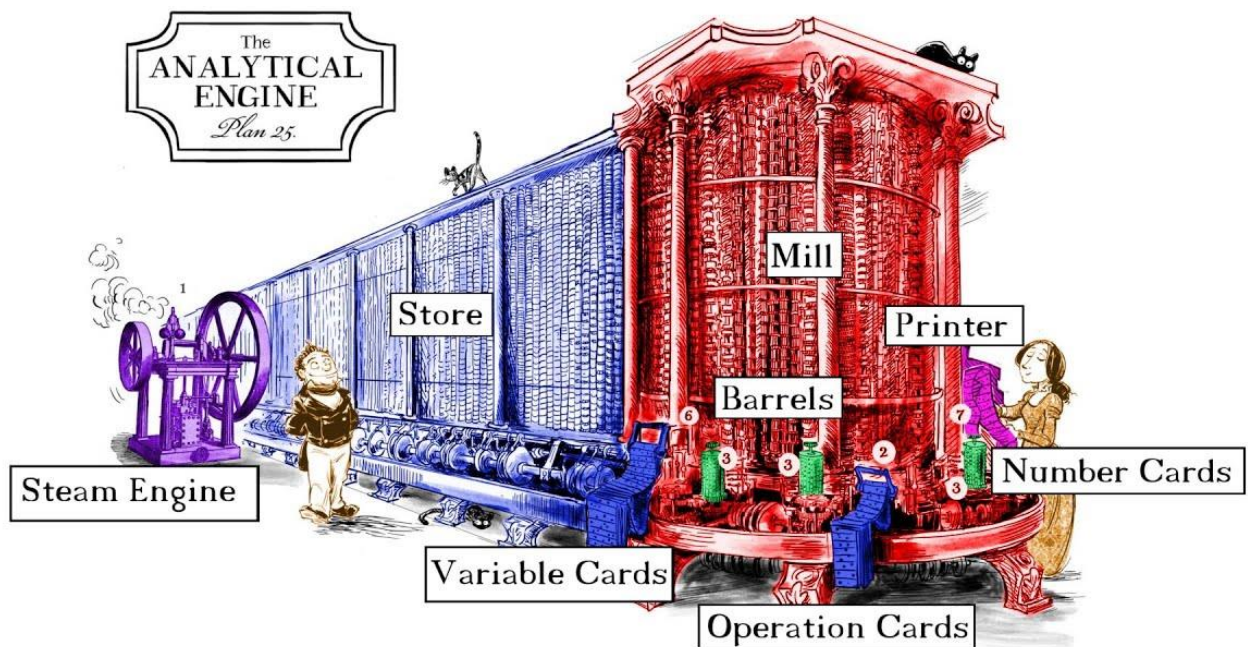
The reader was another new feature of the Analytical Engine. Data (numbers)—as well as instructions for manipulating the data—were to be entered on punched cards. Babbage got this idea from Joseph-Marie Jacquard and his Jacquard loom. The use of instruction cards would make it a programmable device and far more flexible than any machine then in existence. The machine would also be able to carry out instructions in other than sequential order. It was to have a kind of decision-making ability enabling it to jump to a different instruction depending on the data. This extremely powerful feature was missing in many of the early computers of the 20th century.

Babbage ran into difficulty in getting the Analytical Engine built. Many judged his ambitious design unrealistic given the technology of the day. In addition, government funding stopped after Babbage failed to complete his Difference Engine. Even so, the Analytical Engine, although not actually completed, was the first machine that deserved to be called a computer.

Here are the key aspects of the Analytical Engine:

- **General-Purpose Design:** Unlike Babbage's earlier Difference Engine, which was specialized for calculating mathematical tables, the Analytical Engine was designed to perform a wide range of tasks, making it the earliest known design for a general-purpose computer.
- **Mechanical Operation:** Similar to the Difference Engine, the Analytical Engine was envisioned to operate using mechanical components such as gears, shafts, and levers. It would have been powered by steam or other mechanical means.
- **Programmability:** One of the most innovative features of the Analytical Engine was its concept of programmability. Babbage proposed a system where the machine could be instructed to perform different tasks by feeding it a series of instructions, or programs, encoded on punched cards or other media. This concept of programmability laid the foundation for modern computers.
- **Memory:** The Analytical Engine included a memory unit called the "store," which could hold both data and instructions. This allowed for the manipulation and storage of intermediate results during calculations, a crucial feature for general-purpose computing.
- **Arithmetic Logic Unit (ALU) and Control Unit:** The Analytical Engine featured an Arithmetic Logic Unit (ALU) capable of performing arithmetic operations and logical comparisons. It also had a Control Unit responsible for executing instructions and controlling the flow of data within the machine.

- **Printing Mechanism:** Babbage proposed a printing mechanism that would allow the Analytical Engine to automatically print results onto paper, facilitating the output of computations.
- **Unrealized Project:** Despite Babbage's extensive work on the design and theoretical principles of the Analytical Engine, the project was never completed during his lifetime. Technical challenges, funding issues, and other factors prevented its construction.
- **Legacy and Influence:** Although the Analytical Engine was never built, its design and concepts had a profound influence on the development of computing technology. It introduced key concepts such as programmability, memory storage, and automatic control that would later become fundamental components of digital computers.



Mechanical tabulating machine (1890)

The mechanical tabulating machine, also known as the punched card tabulator, was created by Herman Hollerith, an American statistician, and inventor who made significant contributions to the development of data processing equipment. In 1890, he developed a mechanical tabulating machine that utilized punched cards to record and process data. This innovation was initially created to assist with the United States Census of 1890.

While working as an assistant in the Census Office, Hollerith recognized the inefficiencies in data processing and began exploring ways to automate the tabulation process. He was inspired by the Jacquard loom, which used punched cards to control the weaving of patterns in textiles.

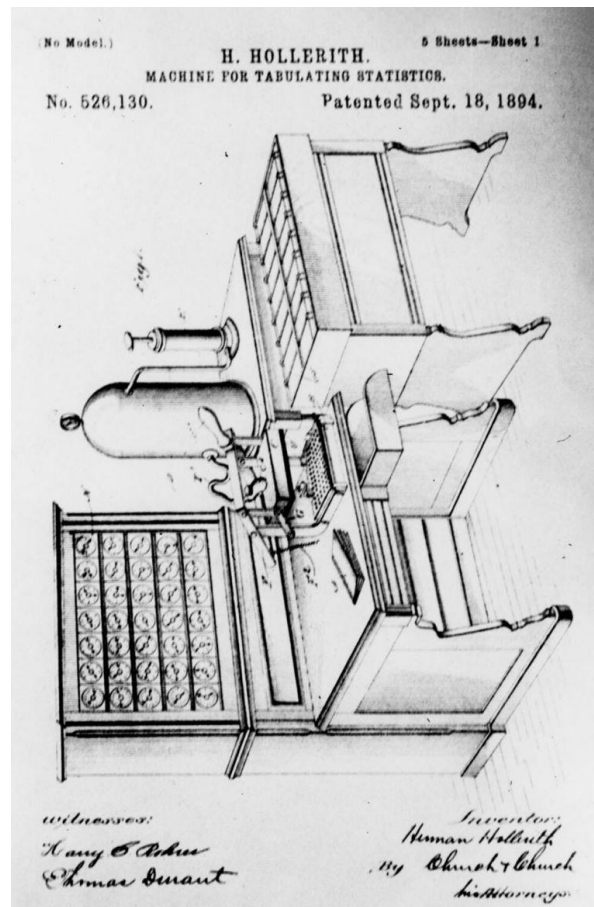
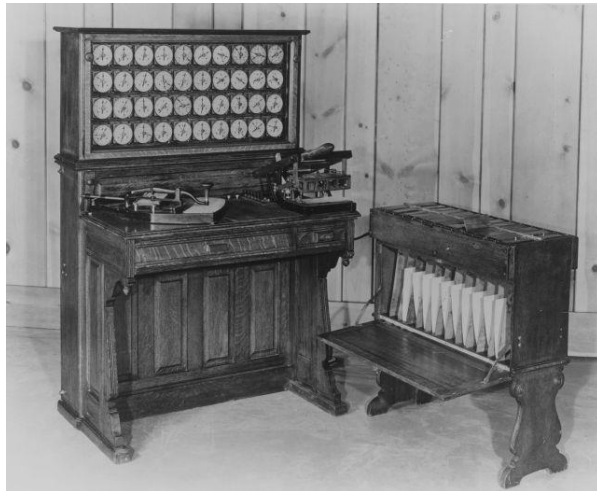
Hollerith's machine used punched cards to represent data. Each card could hold information through the presence or absence of holes in predefined positions. The machine included a card reader that used electrical contacts to detect the holes, and counters that recorded the data.

Encouraged by the success of his invention, Hollerith founded the Tabulating Machine Company in 1896. This company would eventually merge with other companies to form the Computing-Tabulating-Recording Company (CTR) in 1911, which was renamed International Business Machines (IBM) in 1924.

Here are the key features and functions of Hollerith's Mechanical Tabulating Machine:

- **Punched Card Input:** Hollerith's machine utilized punched cards as a means of inputting and storing data. Each punched card represented an individual or data point, with holes punched in specific positions to encode information such as demographic data or census responses.
- **Mechanical Operation:** The tabulating machine operated through a series of mechanical mechanisms, including card readers, counters, and sorting devices. Punched cards were fed into the machine, and mechanical components interpreted the holes to perform various calculations or tabulations.
- **Data Processing:** The primary function of Hollerith's tabulating machine was to count, sort, and tabulate data according to predefined criteria. It could tally the number of individuals in different demographic categories, compile census data, or process other statistical information.

- **Electric Components:** While primarily mechanical in nature, Hollerith's tabulating machine incorporated some electric components, particularly in the card reading mechanism. Electrical contacts were used to detect the presence or absence of holes in the punched cards.
- **Efficiency and Accuracy:** By automating data processing tasks, Hollerith's tabulating machine improved the efficiency and accuracy of tabulation compared to manual methods. It reduced the time and labor required for tasks that would otherwise be time-consuming and error-prone.



3. Electro Mechanical Era – 1840 – 1940

Electronic valve (vacuum tube) invented by Lee De Forest in 1906.

The Harvard Mark I, also known as the IBM Automatic Sequence Controlled Calculator (ASCC), was one of the earliest automatic, programmable electromechanical computers. It was developed by Howard Aiken and his team at Harvard University, in collaboration with IBM, and became operational in 1944.

The development of the Harvard Mark I began in 1937 when Howard Aiken first proposed the concept. He collaborated with IBM, which provided the necessary resources and engineering expertise to build the machine. The actual construction of the Harvard Mark I started in 1939, and the project was completed and the machine officially introduced in 1944.

The Mark I was a large-scale electromechanical computer that used electrical switches, gears, shafts, and clutches to perform calculations. It was capable of automatically executing sequences of instructions, making it programmable. The instructions were input via punched paper tape, and the machine could perform a variety of mathematical operations, including addition, subtraction, multiplication, and division, on large numbers.

The Mark I was primarily designed for scientific and engineering applications, particularly for calculating mathematical tables and solving differential equations. While it was not as fast or versatile as later electronic computers, the Mark I represented a significant advancement in computing technology and played a crucial role in advancing computational capabilities during and after World War II.

Some of the electromechanical devices used in the Mark I included (Additional to Vacuum Tubes):

- **Relays:** Electromechanical relays were used for switching and control purposes within the Mark I. These relays allowed for the control of electrical circuits based on mechanical motion.
- **Rotary Switches:** The Mark I used rotary switches to input data and control the operation of the computer. These switches allowed for the manual input of instructions and data.
- **Shafts and Gears:** The Mark I employed a complex system of shafts and gears to perform arithmetic operations. These mechanical components facilitated the movement and manipulation of data within the computer.

- **Clutches and Couplings:** Clutches and couplings were used to synchronize the movement of different components within the Mark I, ensuring proper operation and timing.

The Harvard Mark I computer, also known as the IBM Automatic Sequence Controlled Calculator (ASCC), had several notable features and specialties:

- **Automatic:** Its "automatic" feature enabled it to execute sequences of instructions without constant human intervention.
- **Electromechanical Design:** The Mark I was an electromechanical computer, meaning it used a combination of electrical and mechanical components to perform calculations. It utilized over 750,000 electromechanical parts, including switches, relays, gears, and shafts.
- **Large-Scale Calculation:** One of the primary purposes of the Mark I was to perform large-scale mathematical calculations, particularly for scientific and engineering applications. It was capable of performing addition, subtraction, multiplication, and division operations on numbers up to 23 digits long.
- **Sequential Operation:** The Mark I operated sequentially, meaning it executed instructions in a predetermined order, one at a time. While it could perform complex calculations, its sequential operation made it relatively slow compared to later electronic computers.
- **Paper Tape Input:** Input to the Mark I was provided via paper tape, which contained instructions and data represented as holes punched in the tape. This method allowed programs and data to be entered and stored for processing.
- **Limited Programming Capability:** The Mark I's programming capabilities were limited compared to modern computers. Programs were written in machine language and had to be manually entered using the paper tape input.

The Harvard Mark I was a significant milestone in the history of computing, demonstrating the feasibility of large-scale automatic computation and paving the way for subsequent developments in electronic computing.

Electronic Era (1940 – Present)

The Electronic Era in the history of computing refers to a period marked by the emergence and widespread adoption of electronic digital computers. This era began in the mid-20th century and continues to the present day. Here are some key aspects of the Electronic Era:

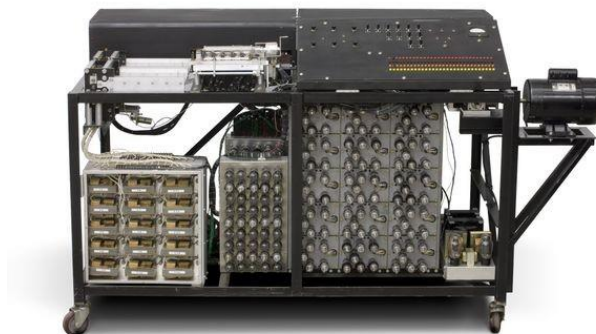
- Introduction of Electronic Components:

The Electronic Era began with the introduction of electronic components such as vacuum tubes, transistors, and later integrated circuits. These components replaced the mechanical switches and relays used in earlier computing devices, enabling faster and more reliable computation.

- Development of Electronic Computers:

The hallmark of the Electronic Era is the development of electronic digital computers.

The first true digital electronic computer was created by John V Atanasoff during 1937-1942. The Atanasoff-Berry Computer was the first to use modern digital switching techniques. Vacuum tubes were used as switches (the needs of switches are explained later). The concepts of using binary arithmetic and logic circuits were introduced to computing world by this ABC.



ABC Computer

The first large-scale electronic computer is generally considered to be the ENIAC (Electronic Numerical Integrator and Computer), completed in 1945 by John Presper Eckert and John William Mauchly. ENIAC was a massive machine that utilized over 18,000 vacuum tubes to perform complex calculations.

- Advancements in Technology:

Throughout the Electronic Era, there were rapid advancements in computer technology driven by innovations in electronic components, circuit design, and manufacturing processes. Vacuum tubes were initially used in early electronic computers, followed by transistors and later integrated circuits, which allowed for smaller, faster, and more powerful computers.

- Commercialization and Expansion: Electronic computers became increasingly commercialized during this era, with companies like IBM, UNIVAC, and DEC (Digital Equipment Corporation) playing key roles in their development and deployment. Computers were used in various industries and applications, ranging from scientific research and military operations to business and government.
- Miniaturization and Personal Computing: Towards the latter half of the 20th century, advancements in integrated circuit technology led to the miniaturization of computers. This facilitated the development of smaller and more affordable computers, paving the way for the personal computer revolution. Companies like Apple, IBM, and Commodore introduced mass-market personal computers, making computing accessible to individuals and small businesses.
- Networking and the Internet: The Electronic Era also witnessed the development of computer networks and the birth of the Internet. ARPANET, the precursor to the modern Internet, was created in the late 1960s by the U.S. Department of Defense. The Internet revolutionized communication and information sharing, ushering in a new era of interconnectedness and digital collaboration.

4. Electronic Era of Computing

1. 1st Generation Computers (1940-1956)

First generation of computer started with using vacuum tubes as the basic components for memory and circuitry for CPU (Central Processing Unit). These tubes like electric bulbs produced a lot of heat and were prone to frequent fusing of the installations, therefore, were very expensive and could be afforded only by very large organizations.

In this generation, punched cards were the predominant method for input, process, output, and data storage.

The Atanasoff-Berry Computer (ABC) is often regarded as the first computer to use vacuum tubes. Developed by John Atanasoff and Clifford Berry at Iowa State College (now Iowa State University) between 1937 and 1942, the ABC employed vacuum tubes in its design for electronic computation. It utilized approximately 300 vacuum tubes as part of its electronic circuitry. While the ABC's impact was not as widely recognized at the time compared to later machines like ENIAC.

The Electronic Numerical Integrator and Computer, also known as the **ENIAC**, consider as the first large-scale computer that used the vacuum tube. It consisted of nearly 20,000 vacuum tubes, as well as 10,000 capacitors and 70,000 resistors. It weighed over 30 tons and took up a lot of space, requiring a large room to house it.

There were machine code and electric wired board languages used.

Used magnetic drums for primary memory*.

The main features of 1st Generation are:

- Vacuum tube technology
- Unreliable
- Supported Machine language only
- Very costly
- Generate lot of heat
- Slow Input/output device
- Huge size
- Non portable
- Consumed lot of electricity



Example Computer in 1st Generation.

1. Atanasoff-Berry Computer (ABC):

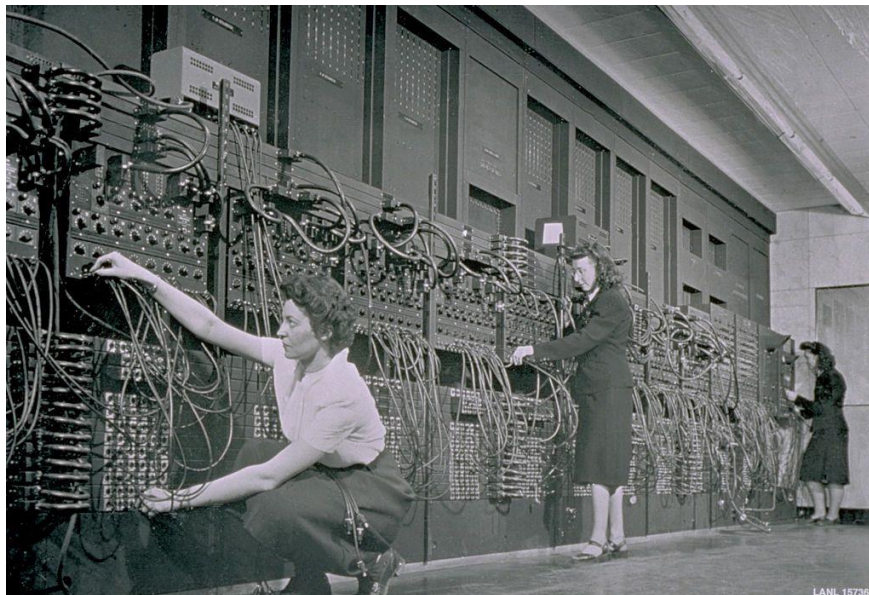
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The Atanasoff-Berry Computer (ABC) is often considered as the first true digital electronic computers. It employed vacuum tubes in its design for electronic computation and introduced key concepts that influenced subsequent developments in computing technology.

2. ENIAC (Electronic Numerical Integrator and Calculator)

Developed by John Mauchly and J. Presper Eckert, ENIAC was the **first electronic general purpose programmable digital computer**, completed in 1946. It was designed to perform complex numerical calculations and was used primarily for military purposes.

Its programming was done through physical rewiring, where operators had to manually set up the connections between various components to perform specific computations. Each time a new program was to be run, the wiring had to be physically reconfigured, making the process cumbersome and time-consuming compared to the stored-program concept.



3. Manchester Baby (Manchester Small-Scale Experimental Machine):

The Manchester Baby, developed by Frederic C. Williams, Tom Kilburn, and Geoff Tootill, was constructed at the University of Manchester in 1948.

The Manchester Baby is renowned as the world's first stored-program computer. It successfully executed its first program on June 21, 1948, making it a pivotal milestone in the history of computing. The Baby demonstrated the feasibility of storing both data and instructions in the same memory unit, which is a defining characteristic of stored-program computers. This computer was for experimental purpose.

4. EDSAC (Electronic Delay Storage Automatic Calculator)

EDSAC was developed by Maurice Wilkes and his team at the University of Cambridge, with construction starting in 1947 and completion in 1949.

While not the first stored-program computer, EDSAC was the first practical and full-size stored-program computer. It operated on the stored-program concept proposed by John von Neumann and utilized mercury delay lines for memory storage.

5. EDVAC (Electronic Discrete Variable Automatic Computer)

Considered one of the first digital computers capable of storing programs, EDVAC was built in 1948. It

played a significant role in the development of stored-program computing but was preceded by the Manchester Baby, EDSAC, and the Atanasoff-Berry Computer (ABC).

6. UNIVAC (Universal Automatic Computer)

An early electrical computer containing thousands of vacuum tubes, UNIVAC utilized punch cards and switches for input, and punch cards for output and data storage. It was developed by Remington Rand and became one of the first commercially successful computers, used for various applications including business data processing and scientific calculations.

2. 2nd Generation Computers (1956 – 1963)

The second generation of computers represents a significant advancement in computing technology compared to the first generation. This era, spanning from the late 1950s to the mid-1960s, is characterized by several key developments that brought about notable improvements in computer performance, reliability, and usability. Here's an overview of the second generation of computers:

01. **Transistors:** The most defining feature of second-generation computers is the replacement of vacuum tubes with transistors as the primary switching and amplifying components. Transistors are semiconductor devices that perform functions similar to vacuum tubes but are smaller, more reliable, and generate less heat. This transition significantly reduced the size and power consumption of computers while improving their reliability and performance.
02. **Magnetic Core Memory:** Second-generation computers introduced magnetic core memory as the primary form of random-access memory (RAM). Magnetic core memory used small magnetic cores to store binary data, offering faster access times and higher storage densities compared to the earlier drum memory used in first-generation computers. This improvement in memory technology contributed to faster and more efficient data processing.
03. *Batch Processing and Operating Systems: Second-generation computers continued to use batch processing as the dominant mode of operation, where programs and data are processed in groups or batches. However, the development of more sophisticated operating systems during this era improved the efficiency of batch processing by automating tasks such as job scheduling, resource allocation, and input/output management. Examples of operating systems from this period include IBM's OS/360 and Burroughs's MCP.*
04. **High-Level Programming Languages:** Second-generation computers saw the introduction and widespread adoption of high-level programming languages such as COBOL (Common Business-Oriented Language), FORTRAN (Formula Translation), and ALGOL (Algorithmic Language). These languages allowed programmers to write code using more human-readable syntax and abstracted away many low-level details of computer hardware, making software development more accessible and efficient.
05. **Commercialization and Mainframes:** The second generation witnessed the commercialization of computers, with the emergence of mainframe computers designed for large-scale data processing and business applications. Companies like IBM, UNIVAC, and Burroughs played significant roles in developing and marketing mainframe systems tailored for commercial and scientific use. Mainframes offered increased processing power, storage capacity, and reliability compared to

earlier computers, making them indispensable tools for organizations handling large volumes of data.

The main features of 2nd Generation are:

- Use of transistors
- Reliable as compared to First generation computers
- Smaller size as compared to First generation computers
- Magnetic core memory used
- Generate less heat as compared to First generation computers
- Consumed less electricity as compared to First generation computers
- Faster than first generation computers
- Still very costly
- Machine language, Assembly language and high-level programming language like FORTRAN, COBOL were used



TRADIC, IBM 1620, IBM 7094, CDC 1604, CDC 3600, were some computers developed in this generation.

3rd Generation Computers (1964 – 1975)

The computers of third generation used Integrated Circuits (ICs). A single IC contains many transistors, resistors, and capacitors along with the associated circuitry. The I.C was invented by Robert Noyce of Fairchild Semiconductor. This development made computers smaller in size, reliable and efficient.

Keyboards evolved and became more standardized, eventually becoming the primary input device for most computers by the third generation.

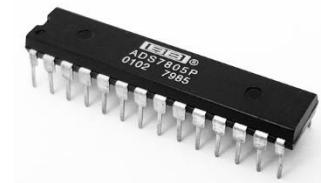
In this generation Remote processing, Time-sharing, Multi-programming Operating System were used.

High level language (FORTRAN-II TO IV, COBOL, PASCAL PL/1, BASIC, ALGOL-68 etc.) were used during this generation.

The first computer to use integrated circuits (ICs) was the IBM 360 Model 85, introduced by IBM in 1965. The IBM 360 Model 85 was a mainframe computer that incorporated small-scale integrated circuits (SSICs) in its design. These SSICs were early versions of integrated circuits, containing a small number of transistors and other components integrated onto a single semiconductor chip.

The main features of 3rd Generation are:

- IC used
- More reliable
- Smaller size
- Generate less heat
- Faster
- Lesser maintenance
- Consumed lesser electricity
- Minicomputers became a common use
- Software industry emerged.
- Support high level language



IBM-360 series, PDP (Personal Data Processor), TDC-316 were the computers of 3rd generation.

4th Generation (1975 – 1989)

The fourth generation of computers is characterized by the widespread adoption of Large Scale Integrated Circuits (LSI) and Very Large Scale Integrated (VLSI) circuits, alongside the development of the microprocessor.

This era witnessed significant advancements in computing technology, resulting in computers that were more powerful, compact, reliable, and affordable. These advancements sparked the personal computer (PC) revolution.

As a result, it gave rise to personal computer (PC) revolution.

In this generation Time sharing, Real time, Networks, Distributed Operating System were used. Specially GUI OS and GUI software application emerged.

High-level programming languages like C, C++, and dBASE were widely used during this era, enabling developers to create sophisticated software applications.

The main features of 4th Generation are

- LSI, VLSI technology used
- Very cheap
- Portable and reliable
- Use of PC's
- Very small size
- Introductions of micro and super computers
- Parallel processing was introduced
- Concept of internet was introduced
- Great developments in the fields of networks
- Computers became easily available

E.g.: DEC 10, STAR 1000, PDP 11

5th Generation Computers (1989 – present)

In the fifth generation, the VLSI technology became ULSI (Ultra Large Scale Integration) technology, resulting in the production of microprocessor chips having ten million electronic components.

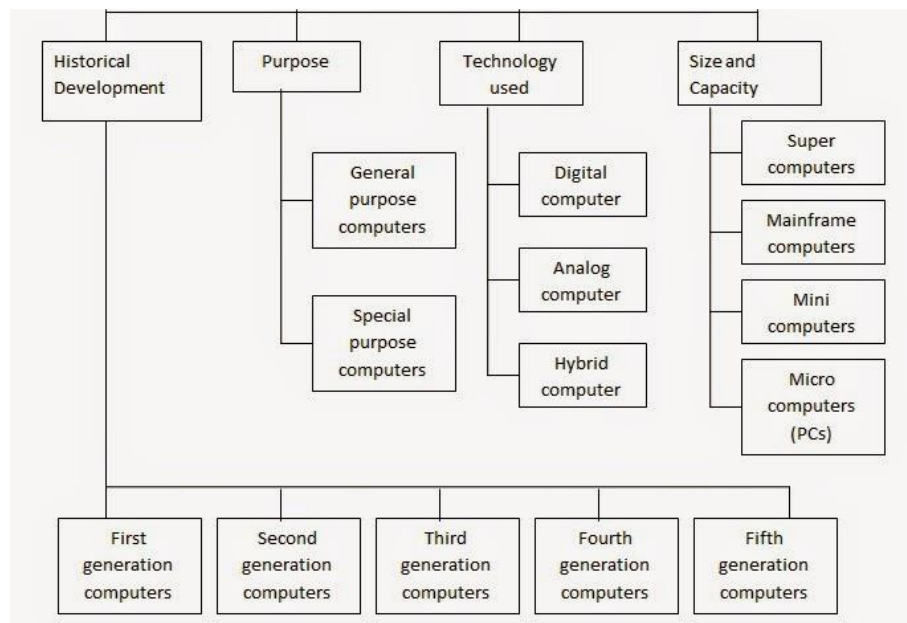
This generation is based on parallel processing hardware and AI (Artificial Intelligence) software. AI is an emerging branch in computer science, which interprets means and method of making computers think like human beings.

The main features of 5th Generation are:

- ULSI technology
- Development of true artificial intelligence
- Development of Natural language processing
- Advancement in Parallel Processing
- Advancement in Superconductor technology
- More user-friendly interfaces with multimedia features
- Availability of very powerful and compact computers at cheaper rates

E.g.: Desktop, Laptop, Notebook, Ultra book

Classification of Computers



1. Based on the Technology

Analog Computer

An analog computer operates using continuous values, such as electrical, mechanical, or hydraulic quantities, to perform computations and solve problems. It processes information by representing physical quantities as continuous signals, making it suitable for tasks involving real-world phenomena and complex simulations.

Digital Computer

A digital computer is designed to process data and perform calculations using discrete values represented as digits, typically in binary format. It operates on discrete signals and employs electronic circuits to manipulate data, enabling precise calculations, logical operations, and storage of information.

Hybrid Computer (Analog + Digital)

A hybrid computer combines the features of both analog and digital computers, allowing it to process both continuous and discrete data. This versatility enables hybrid computers to handle a wide range of applications, from scientific simulations and engineering analysis to real-time control systems. By

leveraging the strengths of both analog and digital technologies, hybrid computers offer a cost-effective solution for tasks requiring complex computations and data processing.

2. Based on the Purpose

Special purpose computer

- Special purpose computers are designed and built to perform specific tasks or functions efficiently and effectively.
- They are optimized for a particular application, industry, or task and typically have hardware and software tailored to meet specific requirements.
- Examples include:

Control systems in industrial automation, such as programmable logic controllers (PLCs) used in manufacturing plants.

Embedded systems in automotive electronics, medical devices, and consumer electronics, where computers are integrated into devices to control specific functions.

Scientific instruments and equipment, such as spectrometers, telescopes, and MRI machines, which use specialized computing capabilities to analyze data and perform measurements.

General purpose computer –

- General purpose computers are versatile machines designed to perform a wide range of tasks and applications.
- They are capable of running different software programs and performing various functions based on user input and requirements.
- Examples include:

Personal computers (PCs) like desktops, laptops, and tablets, which are used for tasks such as web browsing, word processing, multimedia entertainment, and personal productivity.

Workstations used by professionals in fields such as engineering, design, architecture, and multimedia production for more specialized tasks.

Servers that provide services or resources to other computers or users over a network, such as web servers, email servers, and database servers.

3. Based on Size

Super Computer

The fastest and most powerful type of computers Supercomputers are very expensive and are employed. (E.g.: TIANHE-1)

Supercomputers are the most powerful and fastest computers available, designed to handle specialized applications that require immense amounts of computing power.

They are used for applications such as weather forecasting, climate modeling, scientific simulations, nuclear research, and aerospace engineering.

Supercomputers typically feature thousands to millions of processing cores, massive amounts of memory (RAM), and high-speed interconnects for parallel processing.

The unit used to measure and compare the calculation power or processing speed of supercomputers is floating-point operations per second (FLOPS). FLOPS represents the number of floating-point arithmetic operations, such as addition, subtraction, multiplication, and division, that a computer can perform in one second. Supercomputers often achieve performance levels measured in teraflops (trillions of FLOPS) or petaflops (quadrillions of FLOPS), indicating their immense computational capabilities.

Examples include IBM's Summit, NVIDIA's DGX SuperPOD, and Japan's Fugaku, which are among the world's fastest supercomputers.

Mainframe Computer

Mainframe computers are large, powerful, and highly reliable machines used by organizations to process large volumes of data and handle critical business applications. They are capable of supporting thousands of users simultaneously

They are commonly found in industries such as banking, finance, government, healthcare, and telecommunications.

Mini Computer

Minicomputers are mid-sized computers that fall between mainframe computers and microcomputers (personal computers) in terms of processing power and capability.

They were popular in the 1960s to 1980s and were used for tasks such as scientific and engineering calculations, business processing, and network management.

Minicomputers typically offer multi-user and multi-tasking capabilities, making them suitable for shared computing environments.

Micro Computer

Microcomputers, also known as personal computers (PCs), are small, affordable, and widely used computers designed for individual use by consumers, professionals, and students.

- **Desktop:** A personal computer designed to sit on a desk or table, typically consisting of a monitor, keyboard, mouse, and main processing unit. It offers sufficient power and functionality for everyday computing tasks.
- **Laptop:** A portable computer that can be easily carried and used on the go. Laptops typically feature a built-in keyboard, touchpad or trackpad, and display screen, making them convenient for mobile computing.
- **Palmtop:** A handheld computer small enough to fit in the palm of one's hand. Palmtops are compact devices designed for portable computing tasks such as note-taking, scheduling, and basic productivity.
- **Notebook:** A portable computer similar to a laptop but smaller and thinner in design. Notebooks are lightweight and compact, making them ideal for travel and mobile computing while still offering sufficient power for everyday tasks.
- **Smartphone:** A mobile phone equipped with advanced computing capabilities, including an operating system, internet connectivity, multimedia features, and access to mobile apps. Smartphones serve as multifunctional devices for communication, entertainment, and productivity.
- **Tablet:** A wireless, portable computer featuring a touch screen interface and lacking a physical keyboard. Tablets are larger than smartphones but smaller than laptops, offering a balance between mobility and screen size for tasks such as web browsing, media consumption, and light productivity.
- **Phablet:** A hybrid device that combines the features of a smartphone and a tablet, typically featuring a larger screen size than a standard smartphone but smaller than a tablet. Phablets offer a larger display for multimedia and productivity tasks while still retaining the portability of a smartphone. Examples include the iPhone 6 Plus and Samsung Galaxy Note series.